JAKE E. TORRES

MECHANICAL ENGINEER

CONTACT

1+(650) 438-2432

jet2sfo@gmail.com

San Bruno, CA 94066

Los Angeles, CA 90045

PORTFOLIO

WWW. **JAKE-E-TORRES.COM**

Projects | Interests | LinkedIn

EDUCATION

LOYOLA MARYMOUNT UNIVERSITY

AUGUST 2023 - MAY 2027

B.S. Mechanical Engineering

SKILLS

TECHNICAL

- SOLIDWORKS
- AUTOCAD
- MATLAB & SIMULINK
- CFD ANALYSIS
- GD&T
- FSAE
- WELDING
- PROTOTYPING & DESIGN

NON-TECHNICAL

- TEAM COLLABORATION
- PROJECT MANAGEMENT
- LEADERSHIP
- PUBLIC SPEAKING | TEAM COMMUNICATIONS
- CRITICAL THINKING
- CREATIVITY

PROJECTS

Aerodynamics Lead – LMU Formula SAE SEPT. 2024 - MAY 2026

- Led design of side pods, wings, and bodywork to optimize airflow and vehicle performance.
- Modeled components in CAD with focus on manufacturability, tolerancing, and performance goals.
- Conducted CFD analysis to evaluate downforce, drag, and velocity pressure across the aerodynamic package.
- Coordinated with cross-functional sub-teams to ensure seamless component integration.

Wind Turbine Design & Testing DECEMBER 2025

- Designed base, tower, hub, and blade assemblies for a small-scale wind turbine in SolidWorks.
- Optimized geometry for wind tunnel analysis and structural stability.
- Applied NACA 4415 airfoil data to blade design, evaluating lift, power, and efficiency characteristics.
- Integrated a DC motor and evaluated power generation output using a multimeter.

MATLAB & Simulink – 2D Drone Basketball Simulation NOVEMBER 2025

- Modified and tuned multi-axis PID controllers for autonomous quadcopter control.
- Applied root locus analysis to balance drone stability, accuracy, responsiveness, and overshoot.
- Developed state-machine logic to manage ball tracking, electromagnetic actuation, positioning, and shot release.
- Utilized generative AI to assist with code refinement and Simulink control block validation.

LEADERSHIP & ACTIVITIES

President – LMU Formula SAE APRIL 2026 - MAY 2027

- Oversee all Formula SAE sub-teams.
- Lead appointment of an executive board to support technical, financial, and operational functions.
- Coordinate with university administration, faculty advisors, and sponsors on logistics and finances.
- Maintain hands-on involvement in vehicle design, testing, and manufacturing.
- Interpret and enforce FSAE rules and regulations throughout vehicle design and competition preparation.
- Represent the team through networking, sponsor outreach, and competition communications.
- Successfully communicated complex engineering concepts to non-technical audiences.

Sales Associate – Treasure Paws 501(c)(3) MAY 2025 - AUGUST 2025

- Researched and priced donated goods to ensure fair resale aligned with the organization's non-profit mission.
- Supported daily operations through customer service, transactions, and inventory management.
- Performed physical tasks including stock handling and store organization to maintain efficient operations.